Phase 3 & Phase 4 — Plan

Industrial Operations Intelligence Platform — upcoming work (structure + weekly plan).

PHASE 3 — Machine Learning + RAG Document Assistant

Legend: [NEW] = added in this phase | [updated] = changed in this phase | no marker = already exists

Planned folder & file structure (only the parts that change)

```
ops-intelligence-platform/
|
+-- docker-compose.yml                          [updated]
+-- .env.example                                [updated]
|
+-- packages/
|   +-- common/
|       +-- ops_common/
|           +-- domain/
|               +-- registry.py                 [updated]
|
+-- services/
|   +-- ml/                                      [NEW]
|   |   +-- Dockerfile                           [NEW]
|   |   +-- requirements.txt                     [NEW]
|   |   +-- app/
|   |       +-- __init__.py                      [NEW]
|   |       +-- train.py                         [NEW]
|   |       +-- predict.py                       [NEW]
|   |       +-- models/
|   |       |   +-- predictive_maintenance.py    [NEW]
|   |       |   +-- forecasting.py               [NEW]
|   |       |   +-- anomaly_detection.py         [NEW]
|   |       +-- registry.py                      [NEW]
|   |
|   +-- rag/                                      [NEW]
|   |   +-- Dockerfile                           [NEW]
|   |   +-- requirements.txt                     [NEW]
|   |   +-- app/
|   |       +-- __init__.py                      [NEW]
|   |       +-- ingest_docs.py                   [NEW]
|   |       +-- chunker.py                       [NEW]
|   |       +-- embeddings.py                    [NEW]
|   |       +-- vector_store.py                  [NEW]
|   |       +-- retriever.py                     [NEW]
|   |       +-- qa_chain.py                      [NEW]
|   |
|   +-- api-gateway/
|   |   +-- requirements.txt                     [updated]
|   |   +-- api_app/
|   |       +-- routers/
|   |           +-- v1/
|   |               +-- ml.py                    [NEW]
|   |               +-- rag.py                   [NEW]
|   |
|   +-- dashboard/
|       +-- app/
|           +-- api_client.py                    [updated]
|           +-- main.py                          [updated]
|           +-- pages/
|               +-- predictions.py               [NEW]
|               +-- documents.py                 [NEW]
|
+-- data-hub/
|   +-- postgres/
|   |   +-- ml_schema.sql                        [NEW]
|   +-- vector/                                  [NEW]
|       +-- README.md                            [NEW]
|
+-- services/orchestration/
    +-- dags/
        +-- ml_training_dag.py                   [NEW]
```

What we plan to do in the Phase 3 week

- **Predictive maintenance model.** Use the engineered features from Phase 2 to train a model that predicts which machine (entity) is likely to fail or degrade soon.

- **Forecasting model.** Train a time-series model to forecast future values (for example production or demand) from the daily trend data.
- **Anomaly detection.** Flag unusual readings or sudden changes per entity, so problems are caught early.
- **Model training pipeline.** Add an Airflow DAG that trains and refreshes the models automatically on the stored features.
- **Predictions storage + API.** Save model outputs to a new ML schema and expose them through ML API endpoints.
- **RAG document assistant.** Let users upload documents (manuals, reports, logs); chunk them, create embeddings, and store them in a vector database.
- **Question answering.** Build a retriever + QA chain so users can ask questions like "what does this error code mean?" and get answers from their own documents.
- **Dashboard pages.** Add a Predictions page (model results, risk scores, forecasts) and a Documents page (upload docs and chat with them).
- **Keep it grounded.** RAG answers come only from uploaded documents, with sources shown, so the assistant stays accurate and trustworthy.

PHASE 4 — Agentic AI, Copilot & Production Polish

Legend: [NEW] = added in this phase | [updated] = changed in this phase | no marker = already exists

Planned folder & file structure (only the parts that change)

```
ops-intelligence-platform/
|
+-- docker-compose.yml                          [updated]
+-- .env.example                                [updated]
+-- Makefile                                     [updated]
+-- README.md                                    [updated]
|
+-- services/
|   +-- agent/                                    [NEW]
|   |   +-- Dockerfile                          [NEW]
|   |   +-- requirements.txt                    [NEW]
|   |   +-- app/
|   |       +-- __init__.py                     [NEW]
|   |       +-- agent_core.py                   [NEW]
|   |       +-- tools/
|   |       |   +-- query_hub.py                [NEW]
|   |       |   +-- run_analytics.py            [NEW]
|   |       |   +-- call_ml.py                  [NEW]
|   |       |   +-- search_docs.py              [NEW]
|   |       +-- planner.py                      [NEW]
|   |       +-- memory.py                       [NEW]
|   |
|   +-- api-gateway/
|   |   +-- api_app/
|   |       +-- routers/
|   |           +-- v1/
|   |               +-- agent.py                 [NEW]
|   |               +-- auth.py                   [NEW]
|   |
|   +-- dashboard/
|       +-- app/
|           +-- main.py                           [updated]
|           +-- pages/
|               +-- copilot.py                  [NEW]
|               +-- executive.py                [NEW]
|
+-- packages/
|   +-- common/
|       +-- ops_common/
|           +-- auth.py                         [NEW]
|           +-- db.py                           [updated]
|
+-- data-hub/
|   +-- postgres/
|       +-- migrations/                         [NEW]
|           +-- env.py                          [NEW]
|           +-- versions/                       [NEW]
|
+-- tests/                                       [NEW]
|   +-- test_ingestion.py                       [NEW]
|   +-- test_analytics.py                       [NEW]
|   +-- test_api.py                             [NEW]
|
+-- docs/
    +-- demo-script.md                          [NEW]
    +-- architecture.md                         [NEW]
```

What we plan to do in the Phase 4 week

- **Agentic AI core.** Build an agent that can investigate problems on its own — for example, find the root cause of a drop in production by checking multiple domains.
- **Agent tools.** Give the agent tools to query the hub, run analytics, call the ML models, and search documents, so it can gather evidence and reason step by step.
- **Planner + memory.** Add a planner so the agent breaks a question into steps, and memory so it can carry context across a conversation.
- **AI Copilot page.** A chat interface where managers ask questions in plain language and the agent answers using the platform's data, analytics, ML, and documents.
- **Executive dashboard.** A high-level summary view with key KPIs and alerts for decision-makers.

- **Authentication + roles.** Add proper login (JWT) and role-based access, so different users see what they are allowed to.
- **Proper database migrations.** Wire up Alembic so schema changes are versioned and repeatable (deferred from earlier phases).
- **Tests.** Add automated tests for ingestion, analytics, and the API to make the system reliable.
- **Production polish.** One-command startup, clean logging, a README, and a demo script — so the whole platform runs smoothly and is easy to present.